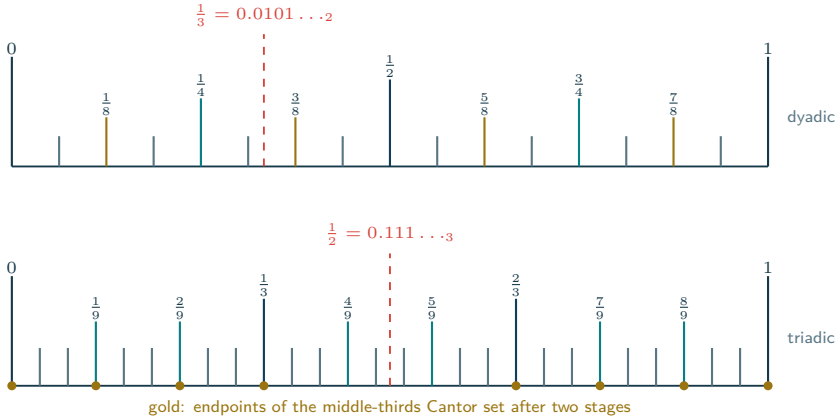
ARITHMETIC ON ONE PAGE

Dyadic and triadic numbers side by side

Halving and thirding give two rulers and two rings, which meet only in the integers and together make $\mathbb{Z}[\frac{1}{6}]$.

1 / THE RULE AND THE PICTURE

Level k adds the marks $m/2^k$ or $m/3^k$ whose numerator is not divisible by the base



gold: endpoints of the middle-thirds Cantor set after two stages

Each ring is $\mathbb{Z}[\frac{1}{p}] = \{m/p^k\}$ for $p = 2$ or 3, and each is closed under adding, subtracting and multiplying but not dividing. The number $\frac{1}{3}$ never lands on the dyadic ruler and $\frac{1}{2}$ never on the triadic one: each has an endless expansion in the other base.

2 / THE COMPARISON

The same construction, with the prime 3 removing less than the prime 2

	dyadic $\mathbb{Z}[\frac{1}{2}]$	triadic $\mathbb{Z}[\frac{1}{3}]$
denominators	2^k	3^k
expansion that stops	binary	ternary
new marks at level k	2^{k-1}	$2 \cdot 3^{k-1}$
marks after k levels	$2^k - 1$	$3^k - 1$
numbers it can divide by	$\pm 2^k$	$\pm 3^k$
a division that leaves the ring	$\frac{1}{2} \big/ \frac{3}{2} = \frac{1}{3}$	$\frac{1}{3} \big/ \frac{2}{3} = \frac{1}{2}$
Euler factor of the inverted prime in $\zeta(2)$	$\frac{1}{1-2^{-2}} = \frac{4}{3}$	$\frac{1}{1-3^{-2}} = \frac{9}{8}$
sum of $1/m^2$ over m coprime to the prime	$\pi^2/8 = 1.2337$	$4\pi^2/27 = 1.4622$

Inverting 2 keeps $\frac{3}{4}$ of $\zeta(2)$ and inverting 3 keeps $\frac{8}{9}$, because a larger prime has a smaller Euler factor, $\frac{p^2}{p^2-1}$.

3 / WHERE THE TWO RINGS MEET

They share only the integers, and together they give everything with denominator $2^a 3^b$

If $m/2^j = n/3^k$ then $3^k m = 2^j n$, which forces both sides to be whole, so $\mathbb{Z}[\frac{1}{2}] \cap \mathbb{Z}[\frac{1}{3}] = \mathbb{Z}$. Since 2^a and 3^b have no common factor, $2^a x + 3^b y = 1$ has whole solutions and $\frac{1}{2^a 3^b} = \frac{y}{2^a} + \frac{x}{3^b}$, so

$$\mathbb{Z}[\tfrac{1}{2}] + \mathbb{Z}[\tfrac{1}{3}] = \mathbb{Z}[\tfrac{1}{6}]$$

For example $\frac{1}{6} = \frac{1}{2} - \frac{1}{3}$ and $\frac{1}{12} = \frac{1}{3} - \frac{1}{4}$. In base 6 both $\frac{1}{2} = 0.3_6$ and $\frac{1}{3} = 0.2_6$ stop, and $\frac{1}{12} = 0.03_6$, so $\mathbb{Z}[\frac{1}{6}]$ is exactly the numbers with a stopping base-6 expansion.

4 / A GAP THE DYADIC RULER DOES NOT HAVE

The triadic marks $\frac{1}{3}, \frac{2}{3}$ bound the first gap of the Cantor set

Delete the open middle third of $[0, 1]$ and repeat. The endpoints after two stages are $0, \frac{1}{9}, \frac{2}{9}, \frac{1}{3}, \frac{2}{3}, \frac{7}{9}, \frac{8}{9}, 1$, all triadic, and the points of the set are the numbers with a ternary expansion using only the digits 0 and 2, for example $\frac{1}{3} = 0.0222\ldots_3$ and $\frac{2}{9} = 0.02_3$.
Both rulers are dense in $[0, 1]$, but only the triadic one carries this construction, whose surviving set has length 0 and still holds uncountably many points.

5 / BOTH PRIMES IN $\zeta(2)$

Together 2 and 3 remove the factor $\frac{4}{3} \cdot \frac{9}{8} = \frac{3}{2}$

Since $\frac{4}{3} \cdot \frac{9}{8} = \frac{3}{2}$ we get $\zeta(2) = \frac{3}{2} \cdot \frac{\pi^2}{9}$, where $\frac{\pi^2}{9} = 1.0966$ is the sum over the whole numbers coprime to 6, and $\xi(2) = \frac{1}{\pi} \cdot \frac{3}{2} \cdot \frac{\pi^2}{9} = \frac{\pi}{6}$, the same value as before.

ASIDE / TRY THESE

$\frac{7}{9} = 0.21_3$ and $\frac{1}{9} = 0.01_3$. Write $\frac{1}{12}$ as a dyadic plus a triadic number: $-\frac{1}{4} + \frac{1}{3}$ or $\frac{1}{3} - \frac{1}{4}$.

LINKS TO OTHER SHEETS

The dyadic sheet sets up $\mathbb{Z}[\frac{1}{2}]$ and $\xi(2)$; the adding-a-new-number sheet gives ring and field.

Convention. $\mathbb{Z}[\frac{1}{p}] = \{m/p^k\}$; Euler factor $p^2/(p^2-1)$; 0.3_6 means $3/6$ in base 6; sums over m coprime to p were checked numerically.

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